

2(a)	the (two) plates are <u>vertical</u> (and separated)	B1
	left plate positively charged and right plate negatively charged/earthed or right plate negatively charged and left plate positively charged/earthed	B1
2(b)	$F = Eq$	C1
	$= 1.3 \times 10^4 \times 3.7 \times 10^{-9}$ $= 4.8 \times 10^{-5} \text{ N}$	A1
2(c)	$F^2 = (4.8 \times 10^{-5})^2 + (5.4 \times 10^{-5})^2$ so $F = 7.2 \times 10^{-5} \text{ N}$ or $F = [(4.8 \times 10^{-5})^2 + (5.4 \times 10^{-5})^2]^{0.5}$ so $F = 7.2 \times 10^{-5} \text{ N}$	A1
2(d)	electric force is constant (because field strength/ E is constant)	B1
	weight is constant (and so resultant force constant)	B1
2(e)(i)	$m = 5.4 \times 10^{-5} / 9.81 (= 5.5 \times 10^{-6})$	C1
	$a = 7.2 \times 10^{-5} / (5.5 \times 10^{-6})$ $= 13 \text{ m s}^{-2}$	A1
2(e)(ii)	$v^2 = u^2 + 2as$ $v^2 = 2 \times 13 \times 0.58$	C1
	$v = 3.9 \text{ m s}^{-1}$	A1